

R21UGS531 - Reasoning and Aptitude

PERIODICAL TEST – I

Question Bank

Part A – MCQ (5 X 2 = 10 Marks)

1. $x : 8 :: 12 : 32$ the value of x is

- a) 4 b) 2 c) 3 d) 16

2. The ratio of the number of sides of the square to the number of edges of a cube is

- a) 1 : 2 b) 3 : 2 c) 4 : 1 d) 1 : 3

3. The length and breadth of a board are 4m and 200cm, then find the ratio of the length and breadth

- a) 1 : 2 b) 2 : 1 c) 1 : 50 d) 50 : 1

4. The average of 10 numbers is 23. If each number is increased by 4, what will the new average be?

- a) 92 b) 23 c) 19 d) 27

5. The average of 5 consecutive odd numbers is 57, then the sum of the largest and smallest numbers is

- a) 61 b) 53 c) 114 d) 8

6. Average of even numbers from 2 to 64

- a) 33 b) 32 c) 65 d) 64

7. 8.4% of a is 42. Then the value of a is

- a) 500 b) 100 c) 1000 d) 50

8. If 280 candidates appeared for an examination, 10% of students failed in the examination, then the number of students who passed the examination is

- a) 250 b) 308 c) 28 d) 252

9. What percent of 60 is 72?

- a) 80% b) 120% c) 84% d) 65%

10. A year ago, the father's age was 7 times that of his son's. Now the father's age is the square of the son's age. Their present ages are

- a) 5, 25 b) 6, 36 c) 7, 49 d) 8, 64

11. John age is 3 years less than the age of Thasleem. The age of Thasleem is 18 years, then the ratio of ages of John and Thasleem is

- a) 5 : 6 b) 6 : 5 c) 3 : 6 d) 6 : 3

12. The age difference between P and Q is 8 years; the ratio of their ages is 2 : 3. Then the present age of P is

- a) 16 years b) 24 Years c) 12 Years d) 30 Years

13. Find the simple interest for ₹5000 at 5% per annum for 5 years? 1250

- a) Rs. 2500 b) Rs. 2000 c) Rs. 3500 d) Rs. 1500

14. At what rate of simple interest will a certain sum be doubled in 10 years

- a) 20% b) 8% c) 10% d) 15%

15. If cost price of 7 toys is equal to selling price of 5 toys. The profit percentage is

- a) 20% b) 10% c) 40% d) 25%

PART -B (5 X 2 = 10 Marks)

1. The ratio of 1 hour to 1 day is _____

2. If 20m : 70m = Rs. 8 : Rs. _____

3. Ratio of 5 paise to 25 paise is the same as the ratio of 20 paise to _____

4. If 15% of 25% of 10000 is _____

5. If 12% of 250 litre is the same as _____% of 150 litre.

6. If 48% of 48 = 64% of x, then x is _____

7. A table is selling ₹ 2470 at a loss of 5% then the C.P is _____

8. Ajay's salary is increased by 18% gives ₹ 236. Find his initial salary?

9. A man buys a fan for ₹1000 and sells it at a loss of 15%. What is the selling price of the fan?

10. The average of 6 numbers is 27, and each number is multiplied by 2. Find the average of a new set of numbers?

11. The average of six consecutive even numbers is 31. Find the smallest number?

12. The average of the first 17 odd numbers is _____.

13. At what rate percent per annum will ₹1,500 amount to ₹1,800 in 4 years?

14. In how many years will ₹2,000 amount to ₹2,400 at 10% per annum simple interest?

15. Find the compound interest when the principal is ₹ 5000, the rate of interest is 10% per annum for 2 years.

PART –C (5 X 6 = 30 Marks)

1. A 84 meter rope is cut into 3 pieces are in the ration 4 : 5 : 3. Find the length of each piece respectively?
2. Given ratios are $a : b = 2 : 3$, $b : c = 5 : 2$, $c : d = 1 : 4$. Find (i) $a : b : c : d$ (ii) $b : d$ (iii) $a : c$
3. Two numbers are in the ratio 3 : 5. If 9 be subtracted from each, then they are in the ratio 12 : 23. Find the numbers?
4. The perimeter of a rectangle is 64 cm. If the ratio of the lengths of two adjacent sides is 7: 9, find the lengths of these sides.
5. In a school there are 800 students and 25 teachers. If the students is increased to 960. How many more teachers should be appointed to maintain the student–teacher ratio?
6. A class of 25 students has an average height of 150 cm. If 5 new students with an average height of 155 cm join the class, what is the new average height of the class?
7. The average of the two classes of students is 50 and 80, respectively. If the first class has 40 students and the second class has 15 students, find the combined average.
8. In the first 12 overs of a cricket match, the run rate was 5.5. What should be the run rate in the remaining 38 overs to reach the target of 300 runs?
9. The average weight of 8 students is 75 kg. If one of the students' weights is 90 kg. What is the average weight of the remaining 7 students?
10. Akila scored 80% of the marks in an examination. If her score was 576 marks, then find the maximum marks of the examination.
11. When 60 is subtracted from 60% of a number to give 60, find the number.
12. In a school of 1400 students, there are 420 girls. Find the number of boys in the school and also find the percentage of boys in the school.
13. Mr. Sidhana saves 28% of his income. If he saves as ₹840 per month, find his monthly income.
14. One year ago, the ratio of Rohan and Pranav ages was 2: 3 respectively. After five years from now, this ratio becomes 4: 5. How old is Pranav now?
15. The ratio of the present ages of P and Q is 4:5. Five years ago, the ratio of their ages was 7:9. Find their present ages. (In years) and also find the sum of their present ages?
16. A father is currently three times as old as his son. Five years ago, he was four times as old as his son. Find the current ages of the father and the son.
17. Ten years ago, the age of mother was three times the age of her son. After ten years, the mother's age will be twice that of his son. Find the ratio of their present ages.
18. If the selling price of an LED TV is equal to $\frac{5}{4}$ of its cost price, then find the gain percentage.

19. By selling a bicycle for ₹4275, a shopkeeper loses 5% for how much should he sell it to have a profit of 5%?
20. If the profit earned on selling an article for ₹810 is the same as loss on selling it for ₹530, then find the cost price of the article.
21. (i) The C.P of 16 boxes of strawberries is equal to the S.P of 20 boxes of strawberries. Find the gain or loss percentage. (ii) Find the difference in C.I and S.I for the principal ₹5000, at the rate of interest 4% p.a. for 2 years.
22. Gayathri invested a sum of ₹12000 at 5% p.a. at compound interest. She received an amount of ₹13230, after 'n' years. Find the value of 'n'.
23. Find the difference in C.I and S.I for the principal ₹8000, at the rate of interest 5% p.a. for 3 years.
24. Find the sum that amounts to ₹2662 at 10% p.a. in 3 years, compounded yearly.

*** All The Best ***